Zahid Ullah Khan

Electrical and ML Engineer

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Machine Learning Engineer with experience in AI/ML model development, data analytics, and visualization. Hands-on experience in building computer vision pipelines, automated workflows, interactive dashboards. Adept at prototype development, exploratory data analysis, and presenting findings clearly to technical and non-technical stakeholders. Passionate about applying computational and AI techniques to interdisciplinary researches.

PROFESSIONAL EXPERIENCE

Airloop — **Islamabad, Pakistan**

Machine Learning Engineer — ***Aug 2022 – Present***

- Optimized the dataset by cleaning, organizing and managing its Meta Data and its sharing within the teams.
- Automating data extraction process to facilitate my team.
- Redesigned and optimized summary reports for our client using data visualization skills.
- Implementing and communicating daily workflow procedures to fellow team members, resulting in an increase in productivity and work flow optimization.
- Training and evaluating different models.
- Orchestrated the collection and analysis of model results, developing automated workflows to compile findings into organized, data-driven reports that bridged the gap between raw metrics and stakeholder interpretation.
- Migrating processed CSV, JSONs and XML data into ArcGIS Pro, creating structured File Geodatabases (GDBs) to enable the development of interactive maps and linked visualizations for object tracking and spatial analysis.
- Developed customed Power BI dashboard for the within teams' meetings for statistical analysis.
- Also developed an application that was used for Data Labeling & Attribute Management of 12 different assets.

Bilal Labs — **Islamabad, Pakistan**

Junior Data Analyst (Remote) — ***Dec 2022 – Oct 2023***

- Assisting team in cleaning raw data and extracting meaningful insights from their data.
- Managing the visualization of data insights and results.

EDUCATION

National University of Sciences and Technology (NUST) — **Islamabad, Pakistan**

MS Information Security — ***2025 – 2026***

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI) — **Swabi, Pakistan**

BS in Electrical Engineering (Power) — ***2018 – 2022***

Cadet College Kohat — **Kohat, Pakistan**

Higher Secondary School Certificate (Pre-Engineering) — ***2015 – 2017***

ACADEMIC PROJECTS

Automated Malaria Parasite Detection in Blood Smears:

- Developed a CNN-based computer vision pipeline to automate the detection of parasitized cells in microscopic blood smear images taken from NIH "malaria cell images" dataset, reducing manual counting time by automating feature extraction and classification.

Optimizing R-CNN for Real-World Crop Disease Identification:

- Engineered a Faster R-CNN model for precision plant disease diagnosis across 14 crop species (PlantVillage dataset), optimizing detection for 38 distinct disease classes.
- Conducted rigorous Exploratory Data Analysis (EDA) on the 54k+ dataset to diagnose severe class imbalances across the 38 categories, implementing statistical re-sampling techniques to prevent model bias toward majority crops. Programming Face Recognition system using MATLAB coding and Data sets.

Data Labeling & Attribute Management Module:

- Developed a core component for a Computer Vision Quality Control (QC) application using React and Electron. This module allows annotators to inspect, edit, and validate metadata for object detection (XML) and segmentation (JSON) datasets.

A particle deep framework based on deep learning for Internet of Things networks:

- Engineered a PSO-optimized MLP model for IoT security, leveraging statistical data analysis (EDA) and feature extraction to resolve class imbalances and automate hyperparameter tuning for maximum accuracy. BOT-IoT dataset was used with 14 most important features.

FYP: Power factor correction (PFC) based induction motor drive for industrial application:

- A real time cost minimizing drive for Induction motors which will be enhancing the power factor, controlling the speed of motor and providing a soft starter for the motor. We used Simulink for the testing the project in initial stages but once we completed the testing and diagnosing stage, we used C++ through Arduino for our final drive.

Junior Year Project: Digital Multi-tasking Energy Meter:

- A multi-function energy meter monitors and measures all the electrical parameters such as input voltage, current, frequency, the power factor, the overall units consumed and their total price of those units. Overall coding was done using Arduino in C++ for all the output requirements from the PIC18f4550 micro-controller

Complex Engineering Problem (CEP) Projects:

- Designed and analyzed a whole distribution system with an effective protection scheme for Norway using MATLAB coding for data extraction and solving matrices.

CERTIFICATIONS

- Coursera Certified: Introduction to Programming with MATLAB.
- Coursera Certified: Getting started with Google Sheets.
- Coursera Certified: Programming for Everybody (Python Data Structures).
- Coursera Certified: Google Data Analytics Professional Certificate.
- DeepLearing.AI Certified: Machine Learning Specialization.
- CISCO certified: Certified Ethical Hacker (CEH)

AWARDS

- Chief Minister KP Scholarship for my Bachelors studies.
- Best Sportsman (2017) and Best Athlete (2015-2017) in college.
- Teams: GIKI Football Team, CCK Football Team, CCK Captain Gymnastic Team, CCK Basketball Team.

SOFTWARE SKILLS

Programming Languages:	Python (Pandas, pytorch, NumPy), SQL, C++, MATLAB, HTML, CSS, React.
Electrical Engineering Tools:	MATLAB, Proteus, Multisim, Power World Simulator, Simulink.
Developer Tools:	Jupyter Lab, Google Colab, VSCode, BigQuery, Tableau, Github, Linux, ArcGIS.
Office Tools:	Google sheets, Libre Office, MS word | Excel | PowerPoint.